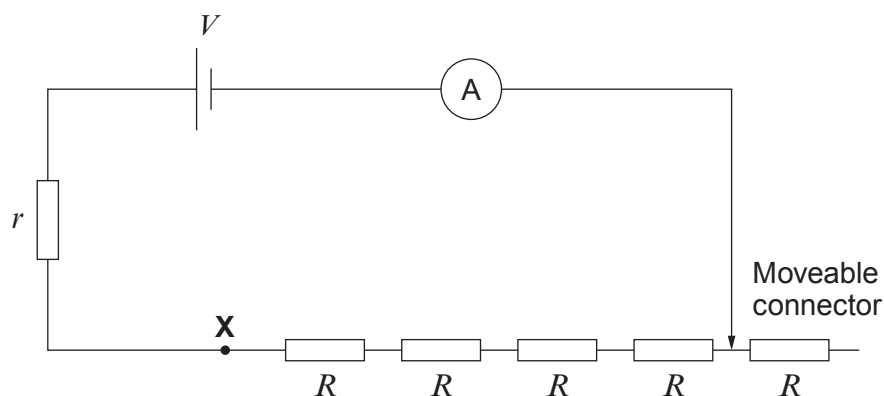


6. A student sets up the following experiment to find how current is related to the number of identical resistors, R , in the circuit. She uses a power supply that provides a constant voltage, V , and includes a safety resistor, r , in the circuit as shown.



- (a) The moveable connector is moved to point X in the circuit. Explain why the ammeter might be damaged if there was no safety resistor, r , included in the circuit. [2]

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- (b) The student takes a series of readings of current as the number of resistors in the circuit is increased. Her results are shown below.

Number of resistors, R	Total circuit resistance (Ω)	Current, I (A)			$\frac{1}{\text{current}} \left(\frac{1}{\text{A}} \right)$
		First reading	Second reading	Mean	
0	10	1.200	1.200	1.200	0.83
1	30	0.390	0.410		2.5
2	50	0.230	0.250	0.240	4.2
4	90	0.133	0.133	0.133	7.5
5	110	0.110	0.108	0.109	9.2



Refer to the table to answer the following questions.

- (i) **Complete** the missing mean value in the table. [1]
Space for calculation.

- (ii) State the value of the resistance of the safety resistor, r . [1]

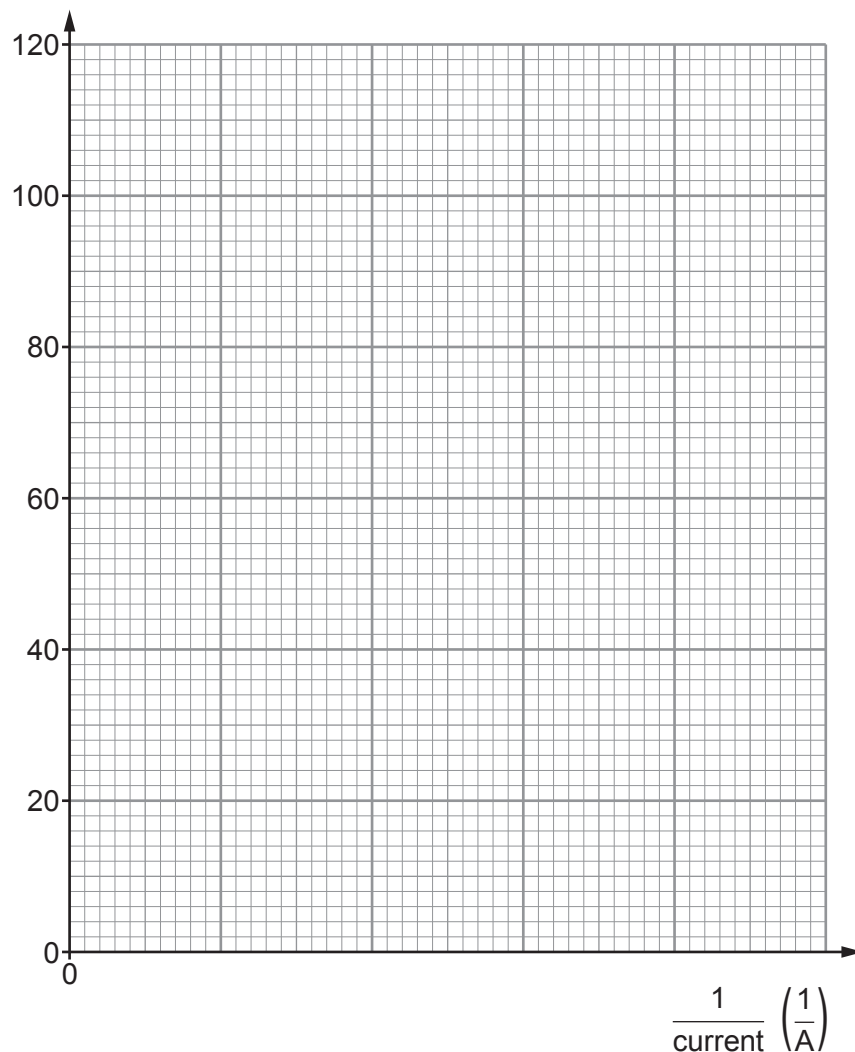
Resistance = Ω

- (iii) Determine the value of the resistance of each resistor, R . [1]

Resistance = Ω

- (iv) Plot the results on the grid below and draw a suitable line. [3]

Total circuit resistance (Ω)



- (c) Use your answer to part (b)(iii) and the graph to determine the current when 3 of the resistors labelled R are included in the circuit. [3]

Current = A

- (d) (i) The gradient of the graph gives the value of the voltage of the battery. Calculate its value and show your workings. [2]

Voltage = V

- (ii) Describe **one** other way that the battery voltage could be calculated. [2]
No calculations are required.

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